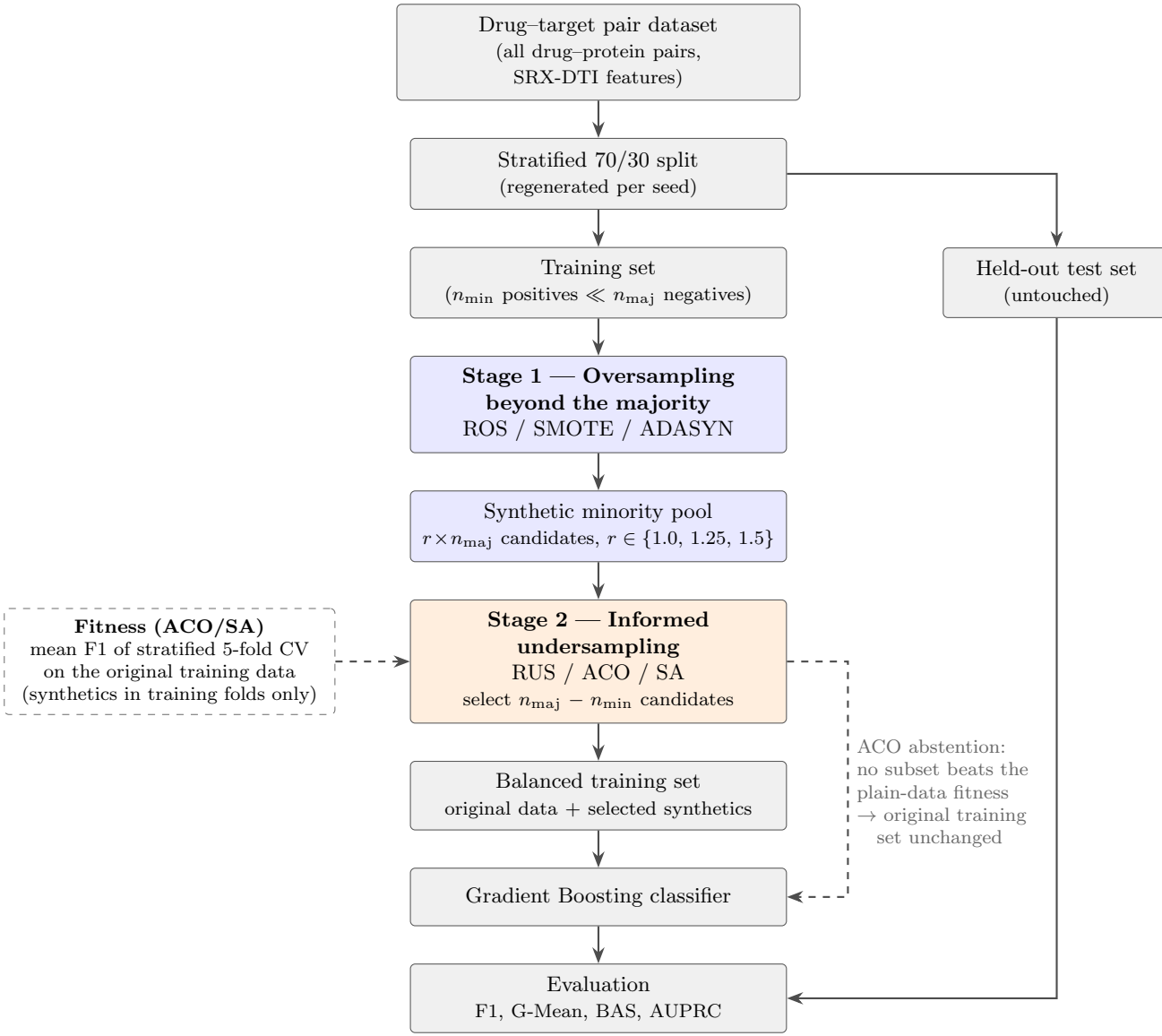

Drug–target pair dataset
(all drug–protein pairs,
SRX-DTI features)
Stratified 70/30 split
(regenerated per seed)
Training set
($n_{\min}$ positives $\ll$ n_{maj} negatives)
Held-out test set
(untouched)
Stage 1 — Oversampling beyond the majority
ROS / SMOTE / ADASYN
Synthetic minority pool
$r \times n_{\mathrm{maj}}$ candidates, $r \in \{1.0, 1.25, 1.5\}$
Fitness (ACO/SA)
mean F1 of stratified 5-fold CV
on the original training data
(synthetics in training folds only)
Stage 2 — Informed undersampling
RUS / ACO / SA
select $n_{\mathrm{maj}} - n_{\min}$ candidates
ACO abstention:
no subset beats the
plain-data fitness
→ original training
set unchanged
Balanced training set
original data + selected synthetics
Gradient Boosting classifier
Evaluation
F1, G-Mean, BAS, AUPRC